

cTheorems

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<https://github.com/sahasatvik/typst-theorems>

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This package provides functions that help create numbered theorem environments, taking heavy inspiration from the L^AT_EX packages `amsthm`, `thmtools`, `apxproof`.

A *theorem environment* combines content with automatically updated *numbering* information. Theorem environments can

- share the same counter (*Theorems* and *Lemmas* often do so)
- have their counters attached to headings or other environments (*Corollaries* are often numbered based upon the parent *Theorem*)
- be `<label>-ed` and `@reference-d`
- be restated or deferred to appear later in the document.

This package also introduces a few miscellaneous features related to mathematical writing.

- Proof environments, with *QED* symbols.
- Equation tags.
- Predefined sets of commonly used theorem environments, and a few themes.

1. Setup

```
#import "@preview/ctheorems:2.0.0": *
#show: thm-rules // Must include!
```

Applying the `thm-rules` show rule — in the same `.typ` file where `ctheorems` functions have been used — is **essential** for displaying theorem environments, references, and equations correctly!

A standard set of theorem environments in the AMS style is available using

```
#import thm-themes.ams: *
```

This lets you immediately use `theorem`, `proof`, `proposition`, `lemma`, `corollary`, `definition`, `example`, `remark`, `claim`. See Appendix B.1 for more details.

2. Demonstration

```
#import "@preview/ctheorems:2.0.0": *
#import thm-themes.ams: *
#show: thm-rules.with(
  qed-symbol:  $\square$ 
)

#definition[Expectation][
  The expectation of a random variable  $X$  on
  a probability space  $(\Omega, \mathcal{E}, \mathbb{P})$  is
  $
  \mathbb{E}[X] = \int X \, d\mathbb{P},
  $
  whenever well-defined.
] <expectation>

#remark[
  We require at least one of  $\mathbb{E}[X^+]$ ,
   $\mathbb{E}[X^-]$  to be finite.
]

#proposition[
  For any  $A \in \mathcal{B}(\mathbb{R})$ ,
   $\mathbb{P}(X \in A) = \mathbb{E}[\mathbf{1}_A(X)]$ .
  $
] <prob-exp>

#theorem[Markov][
  Let  $X \geq 0$ . For all  $a > 0$ ,
   $\mathbb{P}(X > a) \leq \mathbb{E}[X] / a$ .
  $
] <markov>

#corollary[Chebyshev][
  Let  $\mathbb{E}[X] = \mu$ ,  $\text{var}[X] = \sigma^2$ .
  Then,
   $\mathbb{P}(|X - \mu| \geq k\sigma) \leq 1 / k^2$ .
  $
] <chebyshev>

#proof[of @markov][
  $
  \mathbb{P}(X \geq a) \leq \mathbb{E}[\mathbf{1}_{[a, \infty)}(X)]
  \tag*{<@prob-exp>Prop.}<
  \leq \mathbb{E}[(X / a) \mathbf{1}_{[a, \infty)}(X)]
  \leq \mathbb{E}[X] / a. \quad \square
  $
]
```

Definition 2.1 (Expectation). The expectation of a random variable X on a probability space $(\Omega, \mathcal{E}, \mathbb{P})$ is

$$\mathbb{E}[X] = \int X \, d\mathbb{P},$$

whenever well-defined.

Remark. We require at least one of $\mathbb{E}[X^+]$, $\mathbb{E}[X^-]$ to be finite.

Proposition 2.2. For any $A \in \mathcal{B}(\mathbb{R})$,

$$\mathbb{P}(X \in A) = \mathbb{E}[\mathbf{1}_A(X)].$$

Theorem 2.3 (Markov). Let $X \geq 0$. For all $a > 0$,

$$\mathbb{P}(X > a) \leq \frac{\mathbb{E}[X]}{a}.$$

Corollary 2.3.1 (Chebyshev). Let $\mathbb{E}[X] = \mu$, $\text{var}[X] = \sigma^2$. Then,

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Proof of Theorem 2.3.

$$\begin{aligned} \mathbb{P}(X > a) &= \mathbb{E}[\mathbf{1}_{(a, \infty)}(X)] \quad (\text{Prop. 2.2}) \\ &\leq \mathbb{E}\left[\left(\frac{X}{a}\right) \mathbf{1}_{(a, \infty)}(X)\right] \\ &\leq \frac{\mathbb{E}[X]}{a}. \quad \square \end{aligned}$$

3. Features

3.1. Numbered theorems

```
#let theorem = thm.with(
  supplement: "Theorem",
  base-level: 1
)

#theorem[Euclid][
  There are infinitely many primes.
] <euclid>
```

Theorem 3.1 (Euclid). There are infinitely many primes.

Note that `theorem` inherits its numbering from the current heading (the default `thm.base`). By setting `thm.base-level` to `1`, this theorem only uses the first level count from its base.

3.2. References

Since our theorem has been labeled `<euclid>`, we can reference it elsewhere in the document via `@euclid`.

We will supply a proof of `@euclid` later.

We will supply a proof of Theorem 3.1 later.

References can be fully customized through `thm.ref-fmt`. The special supplement `[:]` exposes the name of the theorem environment.

Item `@euclid[:]` is named '`@euclid[:]`'.

Item 3.1 is named '`Euclid`'.

3.3. Chained numbering

Theorem environments can be attached together by setting the child's `thm.base` to the parent's `thm.counter`.

```
#let corollary = thm.with(
  supplement: "Corollary",
  base: "Theorem"
)

#corollary(rewrite: true)[
  There is no largest prime number.
] <largest-prime>

#proof([of @largest-prime], defer: true)[
  The existence of a largest prime would
  imply the finitude of the set of primes.
]
```

Corollary 3.1.1. *There is no largest prime number.*

We have set `thm.restate` and `thm.defer` — go to Section 3.7 to find out more!

3.4. Shared numbering

Theorem environments can share the same numbering by setting a common `thm.counter`.

```
#let proposition = thm.with(
  supplement: "Proposition",
  counter: "Theorem",
  base-level: 1
)

#proposition[
  If a natural number divides  $a$  and  $b$ , it
  also divides  $a - b$ .
]
```

Proposition 3.2. *If a natural number divides a and b , it also divides $a - b$.*

For directly manipulating theorem counters, see `thm-counter-get`, `thm-counter-step`, `thm-counter-update`.

3.5. Proofs

Theorems go hand in hand with proofs, which are available using `proof`.

```
#proof[of @euclid][
  Suppose to the contrary that $p_1, p_2,
  dots, p_n$ is a finite enumeration of
  all primes. Set $P = p_1 p_2 \dots p_n$.
  Since $P + 1$ is not in our list, it cannot
  be prime. Thus, some prime factor $p_j$
  divides $P + 1$. Since $p_j$ also divides
  $P$, it must divide the difference $(P + 1)
  - P = 1$, a contradiction.
]
```

Proof of Theorem 3.1. Suppose to the contrary that $p_1, p_2, \dots, p_n$ is a finite enumeration of all primes. Set $P = p_1 p_2 \dots p_n$. Since $P + 1$ is not in our list, it cannot be prime. Thus, some prime factor p_j divides $P + 1$. Since p_j also divides P , it must divide the difference $(P + 1) - P = 1$, a contradiction. ■

Proof environments will float a QED symbol (■) at the bottom right by default. The symbol can be customized by setting `thm-rules.qed-symbol`. If your proof ends in a block equation, or a list/enum, you can place `qedhere` to correctly position the QED symbol.

```
#show: thm-rules.with(qed-symbol: $square$)

#theorem[Prime gaps][
  There are arbitrarily long stretches of
  composite numbers.
]
#proof[
  For any $n > 2$, consider $
    n! + 2, quad
    n! + 3, quad \dots, quad
    n! + n. #qedhere
$
]
```

Theorem 3.3 (Prime gaps). *There are arbitrarily long stretches of composite numbers.*

Proof. For any $n > 2$, consider

$$n! + 2, \quad n! + 3, \quad \dots, \quad n! + n. \quad \square$$

Caution: `qedhere` does not play well with numbered equations!

3.6. Equation tags

Tags can be inserted into equations using `tag`. These mimic `\tag` in L^AT_EX, floating to the right by default.

```
$
(a + b)^2
&= a^2 + 2 a b + b^2 \
&\leq 2a^2 + 2b^2 #tag[(AM-GM)] \
&\leq 2 \thinmax{a, b}^2.
$
```

$$\begin{aligned} (a + b)^2 &= a^2 + 2ab + b^2 \\ &\leq 2a^2 + 2b^2 \quad (\text{AM-GM}) \\ &\leq 2 \max\{a, b\}^2. \end{aligned}$$

3.7. Defer, restate, and display

Theorem environments can be restated, or deferred to later in the document by setting `thm.restate`, `thm.defer` as in the example in Section 3.3, then calling `thm-restate`.

```
#import thm-state: thm-restate, thm-display

```

Corollary 3.1.1. *There is no largest prime number.*

Proof of Corollary 3.1.1. The existence of a largest prime would imply the finitude of the set of primes. ■

This is often useful for pushing content to the appendix. For finer control, consider using `thm.restate-keys`.

```
Euclid's Theorem states:


```

Euclid's Theorem states: There are infinitely many primes.

The `thm-display` function is the most powerful method for manipulating theorem environments. The following produces a crude outline of named theorem environments (so far), excluding proofs; see `thm.fmt` for more information about the formatting function `fmt`.

```
#thm-display(
  thm ⇒ thm.name ≠ none and thm.supplement ≠ "Proof", // Filter
  fmt: thm ⇒ { // Format
    let head = [*#thm.supplement~#thm.number*]
    if thm.name ≠ none { head = head + [~(#thm.name)] }
    let page = link(thm.loc, [#thm.loc.position().page]) // Get page number
    [#head~#box(width: 1fr, repeat[.])~#page\ ]
  }
)
```

Definition 2.1 (Expectation)	2
Theorem 2.3 (Markov)	2
Corollary 2.3.1 (Chebyshev)	2
Theorem 3.1 (Euclid)	2
Theorem 3.3 (Prime gaps)	4

Setting `thm-display.final` to `true` would extend this search to the entire document.

3.8. Formatting

The default theorem formatting function, `thm-fmt-block` helps with basic customization of the theorem title and body, along with the surrounding block.

```
#let theorem-standout = theorem.with(
  stroke: 1pt + green,
  fill: green.lighten(95%),
  outset: 0.7em,
  spacing: 1.5em
) // All four arguments are passed to 'block'

#theorem-standout(
  title-fmt: x => smallcaps(strong(x))
)[
  There are infinitely many composite
  numbers.
]
```

THEOREM 3.4. *There are infinitely many composite numbers.*

See `thm.fmt` for more information on how to write your own custom formatting function.

4. Acknowledgements

Thanks to

- [MJHutchinson](#) for suggesting and implementing the `base-level` and `base: none` features,
- [rmolinari](#) for suggesting and implementing the `separator: ...` feature,
- [DVDTSB](#) for contributing
 - the idea of passing named arguments from the theorem directly to the `fmt` function.
 - the `number: ...` override feature.
 - the `title: ...` override feature in `thm-plain`.
- [PgBiel](#) for fixing breaking changes in version updates.
- The authors of the $\text{\LaTeX}$ packages `amsthm`, `thmtools`, `apxproof`.
- The awesome devs of [typst.app](#) for their support.

A. Function documentation

A.1. ctheorems

- `thm()`
- `proof()`
- `tag()`
- `thm-rules()`
- `thm-fmt-block()`
- `proof-body-fmt()`

Variables:

- `qedhere`

`thm`

Numbered theorem environment.

```
#thm(base: none)[Name][
  #lorem(10)
]

#thm(supplement: "Corollary", base: "Theorem")[
  #lorem(7)
]
```

Theorem 1 (Name). *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.*

Corollary 1.1. *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Parameters

```
thm(
  ..args: arguments ,
  body: content ,
  supplement: str content ,
  counter: str auto ,
  base: str none ,
  base-level: int none ,
  name: content ,
  number: content auto none ,
  numbering: str function none ,
  restate: bool ,
  defer: bool ,
  restate-keys: array ,
  fmt: function ,
  ref-fmt: function
) -> figure context
```

..args arguments

Extra arguments. See `thm.name`, `thm.fmt`.

body content

Theorem body.

supplement str or content

Theorem supplement.

Default: "Theorem"

counter `str` or `auto`

Environment counter name. If `auto`, defaults to `thm.supplement`.

```
#thm(  
  supplement: "Proposition",  
  counter: "Theorem",  
  base: none  
)[  
  #lore(7)  
]
```

Proposition 2. *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Default: `auto`

base `str` or `none`

Base counter name, whose numbering prefixes the theorem environment numbering. If `"heading"`, use the heading counter. If `none`, the theorem environment maintains a global count with no prefix.

```
#thm(supplement: "Example", base: "Theorem") [  
  #lore(4)  
]
```

Example 2.1. *Lorem ipsum dolor sit.*

Default: `"heading"`

base-level `int` or `none`

Base level, determining the number of levels of the `base` numbering to use for the theorem environment numbering. If `none`, all levels from the `base` numbering are used. Setting a base level higher than what the `base` provides will introduce zeros for padding.

```
#thm(  
  supplement: "Sub-Sub-Theorem",  
  base: "Theorem",  
  base-level: 2  
)[  
  #lore(12)  
]
```

Sub-Sub-Theorem 2.0.1. *Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor.*

Default: `none`

name `content`

Theorem name. If `auto`, the first positional argument from `thm.args` (if present) is chosen as the theorem environment's name.

```
#thm(name: "Named", base: none) [  
  #lore(4)  
]  
#thm(base: none) [Also Named] [  
  #lore(8)  
]
```

Theorem 3 (Named). *Lorem ipsum dolor sit.*

Theorem 4 (Also Named). *Lorem ipsum dolor sit amet, consectetur adipiscing elit.*

Default: `auto`

number `content` or `auto` or `none`

Theorem number. If `auto` (and `thm.numbering` is not `none`), the number is computed based on `thm.counter`, `thm.base`, `thm.base-level`.

```
#thm(number: $dagger dagger$)[
  #lorem(5)
] <d-dag>

Recall @d-dag.
```

Theorem ††. *Lorem ipsum dolor sit amet.*
Recall Theorem ††.

Default: `auto`

numbering `str` or `function` or `none`

Theorem numbering style. If `none`, numbering is suppressed.

```
#thm(supplement: "Remark", numbering: none)[
  #lorem(5)
]
```

Remark. *Lorem ipsum dolor sit amet.*

Default: `"1.1"`

restate `bool`

Mark for being restated later. See `thm-restate()`.

Default: `false`

defer `bool`

Mark for being deferred to later, without appearing in the current position. See `thm-restate()`.

Default: `false`

restate-keys `array`

Keys used by `thm-restate.keys` for filtering. If `auto`, defaults to an array containing `thm.supplement` and `thm.name` (if present).

Default: `auto`

fmt `function`

Formatting function controlling the appearance of the theorem environment, of the form `thm ⇒ content`. `thm` is a dictionary whose keys are essentially the arguments of `thm()`, with a couple of additions/exceptions.

- The keys `name`, `counter`, `number`, `restate-keys` contain their finally inferred/computed values.
- The key `loc` contains the theorem's location.
- The key `args` contains a dictionary of only the named arguments from `thm.args`.

It is often useful to use `thm.args` to control formatting functions.

```
#let thm-color = thm.with(
  base: none,
  fmt: thm => {
    let color = thm.args.at("color", default:
black)
    [*#thm.supplement~#thm.number.* #text(color,
thm.body)]
  }
)

#thm-color[#lorem(15)]
#thm-color(color: blue)[#lorem(7)]
```

Default: thm-fmt-block

Theorem 5. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore.

Theorem 6. Lorem ipsum dolor sit amet, consectetur adipiscing.

ref-fmt function

Formatting function for references, with the same form as `thm.fmt`. The `thm` dictionary carries an additional `ref-supplement` key containing the supplement used in the `@cite[ref-supplement]` call.

```
#let thm-ref-name = thm.with(
  base: none,
  ref-fmt: thm => {
    [#thm.name~(#thm.supplement~#link(thm.loc,
thm.number))]
  }
)

#thm-ref-name[Gauss][#lorem(17)] <gauss>

By @gauss, ...
```

Theorem 7 (Gauss). Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore.

By Gauss (Theorem 7), ...

Default: thm => {

```
  let supplement = thm.supplement
  if thm.ref-supplement != none { supplement = thm.ref-supplement }
  if thm.ref-supplement == [!] and thm.name != none {
    return [#link(thm.loc, thm.name)]
  }
  if supplement != none and supplement != [] { supplement = [#supplement~] }
  [#supplement#link(thm.loc, thm.number)]
}
```

proof

Proof environment. Identical to `thm()`, with different defaults. Uses the `thm-fmt-block()` formatting function (for `thm.fmt`), with `proof-body-fmt()` (for `thm-fmt-block.body-fmt`).

```
#thm(base: none)[
  #lorem(6)
]
#proof[
  #lorem(3)
]
```

Theorem 8. Lorem ipsum dolor sit amet, consectetur.

Proof. Lorem ipsum dolor. ■

Parameters

`proof()` -> function

tag

Add content which floats to the right in an equation.

```
$
(a + b)^2
&= a^2 + 2 a b + b^2 \
&\leq 2a^2 + 2b^2 #tag[(AM-GM)] \
&\leq 2 \text{thin} \max\{a, b\}^2.
$
```

$$\begin{aligned}(a + b)^2 &= a^2 + 2ab + b^2 \\ &\leq 2a^2 + 2b^2 \quad (\text{AM-GM}) \\ &\leq 2 \max\{a, b\}^2.\end{aligned}$$

Parameters

```
tag(
  t: any,
  move: bool,
  left: bool
) -> content
```

t any

Content to use as tag

move bool

If **false**, tags occupy no horizontal space in the equation layout. Set this to **true** if equation content overlaps the tag. This will have the minor disadvantage of moving the equation off-center to accommodate the tag.

```
Bad:
$
a^2 + 2 a b + b^2
\leq 2a^2 + 2b^2 #tag[(AM-GM)]
$
Moved:
$
a^2 + 2 a b + b^2
\leq 2a^2 + 2b^2 #tag(move: true)[(AM-GM)]
$
```

Bad:

$$a^2 + 2ab + b^2 \leq 2a^2 + 2b^2 \quad (\text{AM-GM})$$

Moved:

$$a^2 + 2ab + b^2 \leq 2a^2 + 2b^2 \quad (\text{AM-GM})$$

Note: Trying to determine this option automatically currently causes layout convergence issues in some cases.

Default: **false**

left bool

Tags float to the right by default. Set **true** to float tag to the left.

```
$
(a + b)^2
&= a^2 + 2a b + b^2 \
&\leq 2a^2 + 2b^2
#tag(left: true)[(AM-GM)]
$
```

$$\begin{aligned}(a + b)^2 &= a^2 + 2ab + b^2 \\ (\text{AM-GM}) \quad &\leq 2a^2 + 2b^2\end{aligned}$$

For combining with `tag.move`, make sure to place `tag()` before the equation content.

```
$
(a + b)^2
&= a^2 + 2a b + b^2 \
#tag(left: true, move: true)[(AM-GM)]
&\leq 2a^2 + 2b^2
$
```

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(AM-GM) \leq 2a^2 + 2b^2$$

Default: `false`

thm-rules

Rules for styling theorem environments, references, proofs, etc. *Must appear at the beginning of the document.*

Parameters

```
thm-rules(
  qed-symbol: content,
  doc
) -> content
```

qed-symbol `content`

Symbol displayed at the end of proofs. See `proof()`, `qedhere`, `proof-body-fmt()`. Use as

```
#show: thm-rules.with(
  qed-symbol: $square$
)

#proof[
  #lorem(5)
  $ integral_0^oo sin(x)/x thin d x = pi/2.
  #qedhere $
]
```

Proof. Lorem ipsum dolor sit amet.

$$\int_0^\infty \frac{\sin(x)}{x} dx = \frac{\pi}{2}. \quad \square$$

Default: `$qed$`

thm-fmt-block

Default formatting function for theorem environments, wrapping everything in a block. Intended for use in `thm.fmt`. Named arguments from `thm.args` can override all of the arguments below; any remaining named arguments are passed to the block. Blocks have `width: 100%` set by default.

```
#thm(
  base: none,
  fmt: thm-fmt-block,
  title-fmt: x => strong(smallcaps(x)),
  separator: [ \ ],
  stroke: (left: 1pt),
  inset: (x: 0.7em, y: 0.2em)
)[Special][
  #lorem(16)
]
```

THEOREM 9 (Special)

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et.

Parameters

```
thm-fmt-block(
  thm: dictionary,
```

```

block-args: dictionary ,
name-fmt: function ,
title-fmt: function ,
body-fmt: function ,
separator: content
) -> content

```

thm dictionary

Theorem data. See [thm.fmt](#) for details.

block-args dictionary

Named arguments for the block.

Default: (:)

name-fmt function

Formatting for the environment name.

Default: $x \Rightarrow [(#x)]$

title-fmt function

Formatting for the environment title (head and number).

Default: strong

body-fmt function

Formatting for the environment body.

Default: emph

separator content

Separator between title and body.

Default: [* . *]

proof-body-fmt

Used as [thm-fmt-block.body-fmt](#) by [proof\(\)](#), for properly styling proofs with a qed symbol inserted at the end of the body. Also see [qedhere](#).

```
#show: thm-rules.with(qed-symbol: $"QED"$)

#proof-body-fmt[#lorem(3)]

#proof-body-fmt[
$
  phi.alt(x) = 1/sqrt(2 pi) e^(-x^2/2) #qedhere
$
]
```

Lorem ipsum dolor. QED

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad \text{QED}$$

Parameters

`proof-body-fmt` (`body`: `content`) -> `content`

body `content`

Proof body.

qedhere `metadata`

If placed in a block equation/enum/list within a proof, place a qed symbol to its right.

```
#proof[
  #lorem(3)
  $ x^2 + y^2 = z^2. #qedhere $
]

#proof[
  + #lorem(4)
  + #lorem(5) #qedhere
]

#proof[
  $
    (a + b)^2 &= (a + b)(a + b) \
                &= a^2 + 2 a b + b^2. #qedhere
  $
]
```

Proof. Lorem ipsum dolor.

$$x^2 + y^2 = z^2. \quad \blacksquare$$

Proof.

1. Lorem ipsum dolor sit.
2. Lorem ipsum dolor sit amet. ■

Proof.

$$\begin{aligned} (a + b)^2 &= (a + b)(a + b) \\ &= a^2 + 2ab + b^2. \end{aligned} \quad \blacksquare$$

A.2. thm-state

Methods for restating, reformatting, filtering, and manipulating theorem environments from elsewhere in the document. Every call to `thm` appends a `thm` data dictionary to the state `thm-stored`, where the keys of `thm` are as described in `thm.fmt`.

Import all functions using

```
#import thm-state: *
```

- `thm-restate()`
- `thm-display()`

Variables:

- `thm-stored`

thm-restate

Displays theorem environments which have been marked to be restated or deferred (by setting `thm.restate`, `thm.defer`), can be filtered. Useful for pushing content to the appendix.

```

#let theorem = thm.with(supplement: "Theorem")
#let lemma = thm.with(
  supplement: "Lemma",
  counter: "Theorem",
)
#let definition = thm.with(
  supplement: "Definition",
  body-fmt: x ⇒ x
)

= Heading

#definition[#lorem(2)]
#lemma[#lorem(8)]

#theorem("Name", restate: true)[#lorem(7)]
#proof(defer: true)[
  #lorem(7)
]

#lemma[#lorem(4)]

= Appendix

#thm-restate()

```

1 Heading

Definition 1.1. Lorem ipsum.

Lemma 1.1. *Lorem ipsum dolor sit amet, consectetur adipiscing elit.*

Theorem 1.2 (Name). *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Lemma 1.3. *Lorem ipsum dolor sit.*

2 Appendix

Theorem 1.2 (Name). *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing. ■

Parameters

```

thm-restate(
  ..keys: str content array function,
  fmt: function auto,
  at: label selector location function auto,
  final: bool,
  all: bool
) -> content

```

..keys `str` or `content` or `array` or `function`

String/content keys, array of keys, or functions used to filter theorem environments. A theorem environment is displayed if it passes *any* of the filters.

If `k` in `keys` is a string/content, theorem environments containing `k` in its array of `restate-keys` will be matched.

```

#let theorem = thm.with(supplement: "Theorem")
#let lemma = thm.with(
  supplement: "Lemma",
  counter: "Theorem",
)

= Heading

#theorem(restate: true)[#lorem(6)]
#lemma(restate: true)[#lorem(4)]
#proof(defer: true)[#lorem(7)]
#lemma(restate: true)[#lorem(3)]

= Restate lemmas/proofs
#thm-restate("Lemma", "Proof")

```

1 Heading

Theorem 1.1. *Lorem ipsum dolor sit amet, consectetur.*

Lemma 1.2. *Lorem ipsum dolor sit.*

Lemma 1.3. *Lorem ipsum dolor.*

2 Restate lemmas/proofs

Lemma 1.2. *Lorem ipsum dolor sit.*

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing. ■

Lemma 1.3. *Lorem ipsum dolor.*

If `k` in `keys` is an array of `strings/content`, theorem environments containing *all* keys from `k` in its array of `restate-keys` will be matched.

= Heading

```
#theorem(
  "Result A",
  restate: true,
  restate-keys: ("Theorem", "Result A")
)[#lorem(6)]
#proof(
  defer: true,
  restate-keys: ("Proof", "Result A")
)[#lorem(7)]
#theorem(restate: true)[#lorem(6)]
#theorem(
  "Result B",
  restate: true,
  restate-keys: ("Theorem", "Result B")
)[#lorem(6)]
#proof(
  defer: true,
  restate-keys: ("Proof", "Result B")
)[#lorem(7)]
```

= Restate Result A

```
#thm-restate("Result A")
```

= Restate theorems tagged Result B

```
#thm-restate(("Theorem", "Result B"))
```

3 Heading

Theorem 3.1 (Result A). *Lorem ipsum dolor sit amet, consectetur.*

Theorem 3.2. *Lorem ipsum dolor sit amet, consectetur.*

Theorem 3.3 (Result B). *Lorem ipsum dolor sit amet, consectetur.*

4 Restate Result A

Theorem 3.1 (Result A). *Lorem ipsum dolor sit amet, consectetur.*

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing. ■

5 Restate theorems tagged Result B

Theorem 3.3 (Result B). *Lorem ipsum dolor sit amet, consectetur.*

If `k` in `keys` is a function, it must be of the form `restate-keys → bool`.

= Heading

```
#theorem(
  restate: true,
  restate-keys: (
    "Theorem", "Unproven claim"
  )
)[#lorem(6)]
#theorem(restate: true)[#lorem(6)]
#lemma(
  "Claim D",
  restate: true,
  restate-keys: ("Lemma", "Claim D")
)[#lorem(6)]
```

= Restate claims

```
#thm-restate(
  keys ⇒ keys.any(
    k ⇒ lower(k).contains("claim")
  )
)
```

6 Heading

Theorem 6.1. *Lorem ipsum dolor sit amet, consectetur.*

Theorem 6.2. *Lorem ipsum dolor sit amet, consectetur.*

Lemma 6.3 (Claim D). *Lorem ipsum dolor sit amet, consectetur.*

7 Restate claims

Theorem 6.1. *Lorem ipsum dolor sit amet, consectetur.*

Lemma 6.3 (Claim D). *Lorem ipsum dolor sit amet, consectetur.*

fmt **function** or **auto**

Formatting function, with the same form as `thm.fmt`. The default `auto` uses the originally supplied formatting function `thm.fmt`. See `thm-display.fmt`.

Default: `auto`

at `label` or `selector` or `location` or `function` or `auto`

Location up to which theorem environments will be displayed. The default `auto` uses the location where `thm-restate()` was called. See `thm-display.at`.

Default: `auto`

final `bool`

If `true`, display environments up to the end of the document. Overrides `thm-restate.at`. See `thm-display.final`.

Default: `false`

all `bool`

If `true`, ignores the `thm`'s `restate` or `defer` state (see `thm.restate`, `thm.defer`). Calling `thm-restate(all: true)` is equivalent to `thm-display()`.

Default: `false`

thm-display

Displays all theorem environments, can be filtered.

Every call to `thm()` appends a `thm` dictionary to `thm-stored`. A `thm` dictionary contains complete information about a theorem environment; its keys are as described in `thm.fmt`. This lets theorems be restated, filtered, and otherwise manipulated elsewhere in a document.

```

#let theorem = thm.with(supplement: "Theorem")
#let lemma = thm.with(
  supplement: "Lemma",
  counter: "Theorem",
)
#let definition = thm.with(
  supplement: "Definition",
  body-fmt: x ⇒ x
)

= Heading <h1>

#theorem("Name")[#lore(7)]
#proof[
  #lore(7)
]

= New heading <h2>

#lemma[#lore(8)]
#definition("Thing")[#lore(2)]
#lemma[#lore(4)]

= Display all

#thm-display()

```

1 Heading

Theorem 1.1 (Name). *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing. ■

2 New heading

Lemma 2.1. *Lorem ipsum dolor sit amet, consectetur adipiscing elit.*

Definition 2.1 (Thing). Lorem ipsum.

Lemma 2.2. *Lorem ipsum dolor sit.*

3 Display all

Theorem 1.1 (Name). *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing. ■

Lemma 2.1. *Lorem ipsum dolor sit amet, consectetur adipiscing elit.*

Definition 2.1 (Thing). Lorem ipsum.

Lemma 2.2. *Lorem ipsum dolor sit.*

Parameters

```

thm-display(
  ..filters: function,
  fmt: function auto,
  at: label selector location function auto,
  final: bool
) -> content

```

..filters **function**

Filtering functions. Each f in `filters` is a function `thm ⇒ bool`. A theorem environment is displayed if it passes *any* of the filters.

= Display only theorems/proofs

```
#thm-display(  
  thm => thm.supplement == "Theorem",  
  thm => thm.supplement == "Proof",  
)
```

= Display if `name` is present

```
#thm-display(  
  thm => thm.name != none  
)
```

4 Display only theorems/proofs

Theorem 1.1 (Name). *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing. ■

5 Display if name is present

Theorem 1.1 (Name). *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Definition 2.1 (Thing). Lorem ipsum.

fmt **function** or **auto**

Formatting function, with the same form as `thm.fmt`. The default **auto** uses the originally supplied formatting function `thm.fmt`.

```
#thm-display(  
  thm => thm.supplement != "Proof",  
  final: true,  
  fmt: thm => {  
    let head = [*#thm.supplement~#thm.number*]  
    if thm.name != none {  
      head = head + [~(#thm.name)]  
    }  
    let page = link(thm.loc,  
  [#thm.loc.position().page])  
    [#head~#box(width: 1fr, repeat[.])~#page\ ]  
  }  
)
```

Theorem 1.1 (Name)	18
Lemma 2.1	18
Definition 2.1 (Thing)	18
Lemma 2.2	18

Setting `final: true` ensures that even if this `thm-display()` call is placed at the beginning of the document, all theorem environments are listed.

Default: **auto**

at **label** or **selector** or **location** or **function** or **auto**

Location up to which theorem environments will be displayed. The default **auto** uses the location where `thm-display()` was called.

= Display up to <h2>

```
#thm-display(at: <h2>)
```

6 Display up to <h2>

Theorem 1.1 (Name). *Lorem ipsum dolor sit amet, consectetur adipiscing.*

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing. ■

Default: `auto`

final `bool`

If `true`, display all theorem environments up to the end of the document. Useful for creating lists of theorems in the beginning of documents, before they've been stated. Overrides `thm-display.at`.

Default: `false`

thm-stored `state`

State containing theorem environment data, as an array of `thm` dictionaries.

A.3. `thm-counter`

Exposes methods for manipulating counters used by theorem environments. Import all functions using

```
#import thm-counter: *
```

- `thm-counter-get()`
- `thm-counter-step()`
- `thm-counter-update()`

Variables:

- `thm-counters`

thm-counter-get

Get theorem counter. Must be wrapped in context. See `thm.counter`.

Parameters

```
thm-counter-get(counter: str) -> array
```

thm-counter-step

Step theorem counter forward, relative to a `base` and `base-level`. Must be wrapped in context. See `thm.counter`, `thm.base`, `thm.base-level`.

Parameters

```
thm-counter-step(  
    counter: str,  
    base: str,  
    base-level: int  
)
```

thm-counter-update

Update theorem counter See `thm.counter`.

Parameters

```
thm-counter-update(  
    counter: str,
```

```
    update: int array function
)
```

thm-counters state

State containing a dictionary of theorem environment counters. Each dictionary key is a counter name, and each entry is a counter value (array of integers).

B. Themes

B.1. thm-themes.ams

This theme defines the following versions of `thm` in the AMS style.

- `thm` in the plain style, intended for theorems, lemmas, corollaries, propositions, conjectures.
- `thm-def` in the definition style, intended for definitions, conditions, problems, examples.
- `thm-rem` in the remark style, intended for remarks, notes, annotations, claims, cases, acknowledgments, conclusions.

These styles have been used to provide the following environments.

- `theorem`, `proposition`, `lemma`, `conjecture`, `definition` sharing the counter "Theorem", with the default base "heading".
- `corollary`, `example` sharing the counter "Sub-Theorem", with base "Theorem".
- `problem` with its own counter "Problem", with the default base.
- `remark`, `claim`, `proof` (un-numbered).

```
#import "@preview/ctheorems:2.0.0": *
#import thm-themes.ams: *
#show: thm-rules

#set heading(numbering: "1.")

= Convergence in Probability

#definition[
  We say that  $\{X_n\}$  converges to  $X$  in
  probability if for every  $\epsilon > 0$ ,
   $\mathbb{P}(|X_n - X| > \epsilon) \rightarrow 0$ 
  as  $n \rightarrow \infty$ .
]
#remark[
  This is denoted  $X_n \xrightarrow{p} X$ .
]
#example[
  Let  $\mathbb{E}[X_n] = 0$  and  $\mathbb{E}[X_n^2] \rightarrow 0$ .
  Then,  $X_n \xrightarrow{p} 0$ .
]
#theorem[Weak Law of Large Numbers][
  Let  $\{X_n\}$  be i.i.d. with  $\mathbb{E}[X_1] = \mu$ ,
   $\mathbb{E}[X_1^2] < \infty$ . Then
   $\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mu$ .
]
#proof[
  #lorem(14)
]
```

1. Convergence in Probability

Definition 1.1. We say that $\{X_n\}$ converges to X in probability if for every $\epsilon > 0$,

$$\mathbb{P}(|X_n - X| > \epsilon) \rightarrow 0$$

as $n \rightarrow \infty$.

Remark. This is denoted $X_n \xrightarrow{p} X$.

Example 1.1.1. Let $\mathbb{E}[X_n] = 0$ and $\mathbb{E}[X_n^2] \rightarrow 0$. Then, $X_n \xrightarrow{p} 0$.

Theorem 1.2 (Weak Law of Large Numbers).
Let $\{X_n\}$ be i.i.d. with $\mathbb{E}[X_1] = \mu$, $\mathbb{E}[X_1^2] < \infty$.
Then

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mu.$$

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut. ■

B.2. thm-themes.stripe

This theme creates theorem styles and environments just like `thm-themes.ams`, but with a formatting function `thm-fmt-stripe` which produces colorful stripes on the left sides of theorems.

```
#import "@preview/ctheorems:2.0.0": *
#import thm-themes.stripe: *
#show: thm-rules

#set heading(numbering: "1.")

= Convergence in Probability

#definition[
  We say that  $\{X_n\}$  converges to  $X$  in
  probability if for every  $\epsilon > 0$ ,
   $\mathbb{P}(|X_n - X| > \epsilon) \rightarrow 0$ 
  as  $n \rightarrow \infty$ .
]
#remark[
  This is denoted  $X_n \xrightarrow{p} X$ .
]
#example[
  Let  $\mathbb{E}[X_n] = 0$  and  $\mathbb{E}[X_n^2] \rightarrow 0$ .
  Then,  $X_n \xrightarrow{p} 0$ .
]
#theorem[Weak Law of Large Numbers][
  Let  $\{X_n\}$  be i.i.d. with  $\mathbb{E}[X_1] = \mu$ ,
   $\mathbb{E}[X_1^2] < \infty$ . Then
   $\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mu$ .
]
#proof[
  #lorem(14)
]

// Customize stripe
#theorem(stripe: 2pt + red)[Strong Law of Large
Numbers][
  Let  $\{X_n\}$  be i.i.d. with finite  $\mathbb{E}[X_1] = \mu$ . Then
   $\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\text{a.s.}} \mu$ .
]
```

1. Convergence in Probability

Definition 1.1. We say that $\{X_n\}$ converges to X in probability if for every $\epsilon > 0$,

$$\mathbb{P}(|X_n - X| > \epsilon) \rightarrow 0$$

as $n \rightarrow \infty$.

Remark. This is denoted $X_n \xrightarrow{p} X$.

Example 1.1.1. Let $\mathbb{E}[X_n] = 0$ and $\mathbb{E}[X_n^2] \rightarrow 0$. Then, $X_n \xrightarrow{p} 0$.

Theorem 1.2 (Weak Law of Large Numbers). Let $\{X_n\}$ be i.i.d. with $\mathbb{E}[X_1] = \mu$, $\mathbb{E}[X_1^2] < \infty$. Then

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mu.$$

Proof. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut. ■

Theorem 1.3 (Strong Law of Large Numbers). Let $\{X_n\}$ be i.i.d. with finite $\mathbb{E}[X_1] = \mu$. Then

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\text{a.s.}} \mu.$$

thm-fmt-stripe

Theorem formatting function, producing a colorful stripe on the left side. Wrapper around `thm-fmt-block()`.

Parameters

```
thm-fmt-stripe(
  thm: dictionary,
  ..args: arguments,
  stripe: stroke none,
  inset: relative dictionary
) -> content
```

..args arguments

Arguments, passed on to `thm-fmt-block()`

stripe `stroke` or `none`

Stroke of the stripe.

Default: `2pt + black`

inset `relative` or `dictionary`

Inset for the block.

Default: (x: `1.0em`, y: `0.2em`)